Zonghuan Xu

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Education

Fudan University, Shanghai, China
B.S. in Mathematics and Applied Mathematics, Xianghui Plan, School of Mathematical Sciences, 2024-present
Expected graduation: 2028 | GPA: 3.86 out of 4.00 | Credits completed: 109

Strong Mathematical Background: Earned A/A+ (4.0; A+ is an honors distinction) in nine mathematics courses: Mathematical Analysis A I-II (Honors), Real Analysis (Honors), Numerical Linear Algebra and Optimization (Honors), Mathematical Statistics, Advanced Algebra II, Abstract Algebra, Ordinary Differential Equations, and Elements of Geometry and Topology.

AI Background: Earned A grades in Algorithms and Data Structures (Honors), Artificial Intelligence Fundamentals, Mathematical Foundations of AI, and Natural Language Processing.

English Proficiency: TOEFL iBT 5.5/6.0 (comparable overall score: 110/120), 2026.

University of Toronto, St. George Campus, Toronto, Canada
Incoming Exchange Student, Faculty of Arts & Science, Fall 2026

Research Interests

Machine Learning Theory, Continual Learning, Trustworthy AI, Embodied AI.

Selected Publications and Preprints

Xu, Z. and Ma, X.* *From Order to Distribution: A Spectral Characterization of Forgetting in Continual Learning.* **Preprint, 2026.** arXiv:2604.13460. arXiv

Xu, Z.* and Harish, K. *Optimal Training-Time Scaling in Gradual Adaptation.* **Preprint, 2026.** arXiv:2608.04927. arXiv

Xu, Z.* *Hypothesis Testing with Conditional Queries: Learnability and the Value of Interaction.* **Preprint, 2026.** arXiv:2608.06262. arXiv

Xu, Z., Zheng, X., Wu, Y., and Ma, X.* *When Direct Prediction Fails: Evidence from LLM-Based Misinformation Risk Evaluation.* **Preprint, 2026.** arXiv:2604.06820. arXiv

Xu, Z., Li, J., Zhao, Y., Zheng, X.*, Ma, X.*, and Jiang, Y.-G. *DropVLA: An Action-Level Backdoor Attack on Vision-Language-Action Models.* **Accepted at IROS 2026.** arXiv:2510.10932. arXiv

Li, J., Zhao, Y., Zheng, X., **Xu, Z.**, Li, Y., Ma, X.*, and Jiang, Y.-G.* *AttackVLA: Benchmarking Adversarial and Backdoor Attacks on Vision-Language-Action Models.* **Preprint, 2025.** arXiv:2511.12149. arXiv

* Corresponding author(s).

Selected Research Contributions

Distributional Theory of Forgetting in Continual Learning

Supervisor: Prof. Xingjun Ma | Sole First Author | 2026 | Related manuscript: From Order to Distribution

- Reframed continual-learning forgetting under task distributions; derived an exact operator identity and spectral characterization, yielding the leading asymptotic rate.

Training-Time Scaling in Gradual Adaptation

Research Lead | Sole First & Corresponding Author | 2026 | Related preprint: Optimal Training-Time Scaling in Gradual Adaptation

- Reframed an initial holonomy insight into a training-time optimization problem; developed the main theory and proofs, established s_N^* = Theta(N^-1), designed the experiments, and led the manuscript while mentoring a junior collaborator through technical review and revision.

Conditional-Query Hypothesis Testing

Sole Author | 2026 | Related preprint: Hypothesis Testing with Conditional Queries

- Developed the conditional-query testing framework, established an exact learnability criterion, and proved that adaptivity can reduce the required number of queries by a tight quadratic factor.

Backdoor Safety in VLA Models (code)

Supervisor: Prof. Xingjun Ma | Sole First Author | 2025-2026 | Related publication: DropVLA

- Proposed an action-level backdoor framework for VLA models; built the OpenVLA-OFT and LIBERO poisoning and evaluation pipeline; led the experiments and paper writing.

LLM-Based Misinformation Risk Evaluation

Supervisor: Prof. Xingjun Ma | Sole First Author | 2025-2026 | Related manuscript: When Direct Prediction Fails

- Led a human-grounded study with 317 participants and eight LLM evaluators, showing that credibility scores predicted human sharing better than direct scores and that LLM judgments diverged from human ratings.

Honors & Funding

- Fudan Undergraduate Scholarship (Zhao Yuanpei Student), 2026.
- Chonghua Scholarship, School of Mathematical Sciences, 2026.
- Fudan Undergraduate Basic Disciplines Scholarship, 2024-2025.
- Fudan Undergraduate Research Funding (Junzheng Program), 2025-2026.

Technical Skills

Python, PyTorch, Linux/SSH, Git, Bash, and LaTeX; Transformers/PEFT, W&B, LoRA/OFT fine-tuning; OpenVLA, LIBERO, and RLDS/TFDS; LLM evaluation and human-subject data analysis.

Personal Interests

Badminton (regular technical training and competitive match play); strategy games.